

a) Write the motion and measurement equations for this system and identify the matrices necessary for the Kalman filter: A, B, Q, H, R .

+ Define the state vector: $x = \begin{bmatrix} x \\ y \\ v_x \\ v_y \end{bmatrix}$

+ The system has constant acceleration, and the acceleration is input $u = \begin{bmatrix} a_x \\ a_y \end{bmatrix} = \begin{bmatrix} 30 \\ 20 \end{bmatrix}$

+ Motion equation:

Given state transition given $T = 0.5$ s

$$x_k = Ax_{k-1} + Bu + w_{k-1}$$

$$A = \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} T^2/2 & 0 \\ 0 & T^2/2 \\ T & 0 \\ 0 & T \end{bmatrix} = \begin{bmatrix} 0.125 & 0 \\ 0 & 0.125 \\ 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$

+ Process noise covariance Q :

Given: $\sigma_{a_x}^2 = 5$; $\sigma_{a_y}^2 = 4$

Noise is independent, so: $Q = G \cdot Q_a \cdot G^T$

$$G = \begin{bmatrix} T^2/2 & 0 \\ 0 & T^2/2 \\ T & 0 \\ 0 & T \end{bmatrix} = \begin{bmatrix} 0.125 & 0 \\ 0 & 0.125 \\ 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$

$$Q_a = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}$$

$$G^T = \begin{bmatrix} 0.125 & 0 & 0.5 & 0 \\ 0 & 0.125 & 0 & 0.5 \end{bmatrix}$$

$$\Rightarrow Q = \begin{bmatrix} 0.125 & 0 \\ 0 & 0.125 \\ 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 0.125 & 0 & 0.5 & 0 \\ 0 & 0.125 & 0 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.0781 & 0 & 0.3125 & 0 \\ 0 & 0.0625 & 0 & 0.25 \\ 0.3125 & 0 & 1.25 & 0 \\ 0 & 0.25 & 0 & 1 \end{bmatrix}$$

+ Measurement equation:

Measurement vector:

$$z_k = \begin{bmatrix} x \\ y \end{bmatrix} = Hx_k + v_k$$

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \text{ (since we're measuring position } (x, y) \text{ directly)}$$

+ Measurement noise covariance R :

$$\text{Given } R = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

b/ Apply the Kalman filter to estimate the position and velocity of the object for the two first iterations:

k	1	2
x	5	2
y	20	12

+ k = 1

1/ Prediction:

Predicted state:

$$\hat{x}_{1|0} = A \hat{x}_{0|0} + B u_0 = \begin{bmatrix} 1 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.125 & 0 \\ 0 & 0.125 \\ 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 30 \\ 20 \end{bmatrix} = \begin{bmatrix} 3.75 \\ 2.5 \\ 15 \\ 10 \end{bmatrix}$$

Predicted covariance:

$$P_{1|0} = A P_0 A^T + Q = \begin{bmatrix} 1 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0.5 & 0 & 1 & 0 \\ 0 & 0.5 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0.0781 & 0 & 0.3125 & 0 \\ 0 & 0.0625 & 0 & 0.25 \\ 0.3125 & 0 & 1.25 & 0 \\ 0 & 0.25 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 7.5781 & 0 & 5.3125 & 0 \\ 0 & 7.5625 & 0 & 5.25 \\ 5.3125 & 0 & 11.25 & 0 \\ 0 & 5.25 & 0 & 11 \end{bmatrix}$$

2/ Update:

Measurement at k = 1 ; $z_1 = \begin{bmatrix} 5 \\ 20 \end{bmatrix}$

$$K_1 = P_{1|0} H^T (H P_{1|0} H^T + R)^{-1}$$

$$= \begin{bmatrix} 7.5781 & 0 & 5.3125 & 0 \\ 0 & 7.5625 & 0 & 5.25 \\ 5.3125 & 0 & 11.25 & 0 \\ 0 & 5.25 & 0 & 11 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 7.5781 & 0 & 5.3125 & 0 \\ 0 & 7.5625 & 0 & 5.25 \\ 5.3125 & 0 & 11.25 & 0 \\ 0 & 5.25 & 0 & 11 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1}$$

$$= \begin{bmatrix} 0.7912 & 0 \\ 0 & 0.8832 \\ 0.5547 & 0 \\ 0 & 0.6131 \end{bmatrix}$$

$$\hat{x}_{1|1} = \hat{x}_{1|0} + K_1 (z_1 - H \hat{x}_{1|0}) = \begin{bmatrix} 3.75 \\ 2.5 \\ 15 \\ 10 \end{bmatrix} + \begin{bmatrix} 0.7912 & 0 \\ 0 & 0.8832 \\ 0.5547 & 0 \\ 0 & 0.6131 \end{bmatrix} \left(\begin{bmatrix} 5 \\ 20 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3.75 \\ 2.5 \\ 15 \\ 10 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 4.7390 \\ 17.9562 \\ 15.6933 \\ 20.7299 \end{bmatrix}$$

$$P_{1|1} = (I - K_1 H) P_{1|0} = \left(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.7912 & 0 \\ 0 & 0.8832 \\ 0.5547 & 0 \\ 0 & 0.6131 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \right) \begin{bmatrix} 7.5781 & 0 & 5.3125 & 0 \\ 0 & 7.5625 & 0 & 5.25 \\ 5.3125 & 0 & 11.25 & 0 \\ 0 & 5.25 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 1.5824 & 0 & 1.1093 & 0 \\ 0 & 0.8832 & 0 & 0.6131 \\ 1.1093 & 0 & 8.3034 & 0 \\ 0 & 0.6131 & 0 & 7.7810 \end{bmatrix}$$

+ k = 2

1/ Prediction

$$\hat{x}_{2|1}^- = A\hat{x}_{1|1} + Bu_1 = \begin{bmatrix} 1 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4.7390 \\ 17.9562 \\ 15.6933 \\ 20.7299 \end{bmatrix} + \begin{bmatrix} 0.125 & 0 \\ 0 & 0.125 \\ 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 30 \\ 20 \end{bmatrix} = \begin{bmatrix} 16.3357 \\ 30.8211 \\ 30.6933 \\ 30.7299 \end{bmatrix}$$

$$P_{2|1}^- = AP_{1|1}A^T + Q$$

$$= \begin{bmatrix} 1 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1.5824 & 0 & 1.1093 & 0 \\ 0 & 0.8832 & 0 & 0.6131 \\ 1.1093 & 0 & 8.3034 & 0 \\ 0 & 0.6131 & 0 & 7.7810 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0.5 & 0 & 1 & 0 \\ 0 & 0.5 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0.0781 & 0 & 0.3125 & 0 \\ 0 & 0.0625 & 0 & 0.25 \\ 0.3125 & 0 & 1.25 & 0 \\ 0 & 0.25 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4.8456 & 0 & 5.5735 & 0 \\ 0 & 3.5041 & 0 & 4.7536 \\ 5.5735 & 0 & 9.5534 & 0 \\ 0 & 4.7536 & 0 & 8.7810 \end{bmatrix}$$

2/ Update

$$\text{Measurement at } k=2; z_2 = \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$K_2 = P_{2|1}^- H^T (HP_{2|1}^- H^T + R)^{-1}$$

$$= \begin{bmatrix} 4.8456 & 0 & 5.5735 & 0 \\ 0 & 3.5041 & 0 & 4.7536 \\ 5.5735 & 0 & 9.5534 & 0 \\ 0 & 4.7536 & 0 & 8.7810 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4.8456 & 0 & 5.5735 & 0 \\ 0 & 3.5041 & 0 & 4.7536 \\ 5.5735 & 0 & 9.5534 & 0 \\ 0 & 4.7536 & 0 & 8.7810 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1}$$

$$= \begin{bmatrix} 0.7078 & 0 \\ 0 & 0.7780 \\ 0.8142 & 0 \\ 0 & 1.0554 \end{bmatrix}$$

$$\hat{x}_{2|2} = \hat{x}_{2|1}^- + K_2(z_2 - H\hat{x}_{2|1}^-) = \begin{bmatrix} 16.3357 \\ 30.8211 \\ 30.6933 \\ 30.7299 \end{bmatrix} + \begin{bmatrix} 0.7078 & 0 \\ 0 & 0.7780 \\ 0.8142 & 0 \\ 0 & 1.0554 \end{bmatrix} \left(\begin{bmatrix} 2 \\ 12 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 16.3357 \\ 30.8211 \\ 30.6933 \\ 30.7299 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 6.1883 \\ 16.1787 \\ 19.0216 \\ 10.8660 \end{bmatrix}$$